

Reviewer 1

Novelty: 6 (reasonably novel - there are some notable differences from existing ideas and probably enough to turn into a new paper)

Rationale: Focus on the long-form setting is novel at the moment. The idea of obtaining modular confidence estimates for different claims in a long-form output, and synthesizing them into a single uncertainty estimate is not that complicated, but it does seem to be underexplored.

Feasibility: 8 (Highly Feasible: Straightforward to implement the idea and run all the experiments.)

Rationale: The only part of the project that seems challenging is obtaining correctness annotations for one of the datasets (e.g., Essay Writing). GSM8K and code datasets like HumanEval seem like very natural long-form output settings to try out the idea. Other than this, iterating on the prompts for decomposition / verbalized UQ for each of the modules will be important, but the author mentions this.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)

Rationale: It's possible that first obtaining verbalized uncertainty estimates for each module, and then synthesizing into a single score, will outperform the standard baselines of self-consistency over the entire long-form output (using majority vote as the confidence score). However, I don't expect this to be dramatically better. If the paper instead set out with the goal of actually producing the UQ estimates for each claim, then almost no prior work does this, and the baselines would be less strong.

Excitement: 5 (Leaning negative: it has interesting bits but overall not exciting enough)

Rationale: This seems like the most straightforward possible way to obtain uncertainty estimates for a long-form generation with an LLM. This means the project could produce some useful engineering artifacts, but it doesn't really push the idea to its logical conclusion. Therefore I don't consider it "exciting enough". There is some mention of "using the uncertainty estimates to possibly condition on more information" but this is not fleshed out – it could be more interesting. For example, studying how the fine-grained uncertainty estimates could be used to selectively retrieve factual information from Wikipedia etc. on a knowledge-intensive task.

Overall Score: 5 (Decent idea but has some weaknesses or not exciting enough, marginally below the acceptance threshold of major AI conferences)

Rationale: I like the focus on long-form generations. However, this proposal is a very straightforward baseline and extension of existing work to the long-form generation setting (just produce the long generation, decompose it, apply verbalized uncertainty on each claim, and finally aggregate them). I could see the paper being well-cited, but I don't see an interesting/novel angle here.

Confidence: 5 (You are absolutely certain that the evaluation is correct and very familiar with the relevant literature)

Reviewer 2

Novelty: 6 (reasonably novel - there are some notable differences from existing ideas and probably enough to turn into a new paper)

Rationale: While existing works have explored the problem of calibration in long-form answers (e.g. <https://arxiv.org/abs/2402.06544>), the specific method for calibration is different. Also seems related to FactScore (<https://arxiv.org/abs/2305.14251>) where the task was different (getting a factuality score) but the idea of breaking long form generations into smaller units, evaluating each separately and then combining does seem related.

Feasibility: 8 (Highly Feasible: Straightforward to implement the idea and run all the experiments.)

Rationale: The idea seems simple enough to implement with API access, considering all the steps involved in the method can be done via prompting with API. The proposal does mention using LLaMA3-70B as an additional model, which would require GPUs I guess.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)

Rationale: Since it has been shown that LLMs are quite well calibrated when asked to verbalize the confidence for short answers, I'm guessing the calibration scores would be pretty good for individual modules. Also LLMs might be decent at combining confidence scores (especially with detailed instructions and some examples in the prompt), so overall the method might work well. But it's unclear if it would do better than the methods proposed in - <https://arxiv.org/abs/2402.06544>.

Excitement: 6 (Learning positive: exciting enough to be accepted at a major AI conference, but still has some weaknesses or somewhat incremental)

Rationale: If the method does work well in getting calibration for long-form answers, I think that would be pretty exciting. One thing which is missing from the proposal (and why the score was not higher) was that it does not touch upon the issue that for long-form answers we won't have a binary correct/incorrect decision but answers can be partially correct.

Overall Score: 6 (Marginally above the acceptance threshold of major AI conferences)

Rationale: The overall idea makes sense to me, but the score is not higher right now because: (a) it's unclear what exactly is meant by 'modules' especially for essay writing which the proposal mentions as one of the tasks ; (b) the issue for partial correctness which was mentioned above.

Confidence: 3 (You are fairly confident that the evaluation is correct)

Q Example Idea: Semantic Resonance Uncertainty Quantification

Semantic Resonance Uncertainty Quantification (SRUQ) (Part 1)

1. Problem Statement: Current uncertainty quantification methods for Large Language Models (LLMs) often rely on simple statistical measures or model-specific attributes, which may not capture the nuanced semantic uncertainties in complex reasoning tasks. This limitation can lead to overconfident or poorly calibrated model outputs, potentially resulting in unreliable decision-making in critical applications.

2. Motivation: Existing approaches typically use softmax probabilities, entropy measures, or ensemble disagreement to quantify uncertainty. However, these methods often fail to capture the semantic nuances and reasoning complexities in tasks that require deep understanding and multi-step reasoning. Human experts, on the other hand, gauge their uncertainty by considering how well their reasoning 'resonates' with their broader knowledge and experience. By mimicking this process in LLMs, we can potentially develop a more robust and semantically grounded approach to uncertainty quantification.

3. Proposed Method: We propose Semantic Resonance Uncertainty Quantification (SRUQ), which prompts the LLM to generate multiple independent reasoning paths for a given problem, then quantifies uncertainty based on the semantic coherence and mutual reinforcement among these paths. The process involves five key steps:

1. Generating diverse solution attempts using different prompting strategies.
2. Cross-evaluating each solution attempt against the others, assessing logical consistency and mutual support.
3. Constructing a 'resonance graph' where nodes are solution attempts and edges represent semantic reinforcement.
4. Computing a resonance score based on graph properties like connectivity and centrality.
5. Mapping the resonance score to a calibrated uncertainty estimate.

Semantic Resonance Uncertainty Quantification (SRUQ) (Part 2)

4. Step-by-Step Experiment Plan:

1. Dataset Preparation

- Utilize three datasets covering different reasoning tasks:
 - (a) GSM8K for mathematical problem-solving
 - (b) EntailmentBank for logical deduction
 - (c) HotpotQA for multi-hop question answering
- Split each dataset into train, validation, and test sets if not already done.

2. Baseline Implementation

- Implement three baseline uncertainty quantification methods:
 - (a) Softmax probabilities
 - (b) Monte Carlo Dropout
 - (c) Ensemble disagreement (using different few-shot prompts)
- Generate predictions and uncertainty estimates on the validation and test sets for each baseline.

3. SRUQ Implementation

- (a) Generate 5 diverse solution attempts using different few-shot prompts and temperature settings.
- (b) For each pair of solutions, prompt the LLM to evaluate their consistency and mutual support.
- (c) Construct the resonance graph using the pairwise evaluations.
- (d) Compute the resonance score using graph centrality measures (e.g., PageRank).
- (e) Map the resonance score to a calibrated uncertainty estimate using isotonic regression on the validation set.

4. Evaluation

- Compare SRUQ against the baselines using the following metrics:
 - (a) Expected Calibration Error (ECE)
 - (b) Brier score
 - (c) Area Under the Precision-Recall Curve (AUPRC) for uncertainty ranking
- Evaluate the correlation between uncertainty estimates and actual errors.

5. Analysis

- Visualize the resonance graphs for high and low uncertainty examples.
- Analyze the relationship between graph properties and prediction accuracy.
- Investigate cases where SRUQ significantly outperforms or underperforms compared to baselines.

6. Ablation Studies

- Vary the number of solution attempts.
- Compare different graph centrality measures.
- Evaluate the impact of the cross-evaluation step.

7. Generalization Test

- Test the generalization of SRUQ on out-of-distribution samples by applying the method trained on one dataset to examples from the other datasets.

Semantic Resonance Uncertainty Quantification (SRUQ) (Part 3)

5. Test Case Examples:

- **Baseline Example:**
 - Input: Q: If a train travels at 60 miles per hour, how far will it travel in 2.5 hours?
 - Softmax Output: The train will travel 150 miles in 2.5 hours. (Confidence: 0.92)
 - Explanation: The softmax probability is high, but it does not capture the reasoning process or potential uncertainties in the calculation.
- **SRUQ Example:**
 - Input: Q: If a train travels at 60 miles per hour, how far will it travel in 2.5 hours?
 - Solution Attempts:
 1. Distance = Speed $\times$ Time
 - * Distance = 60 miles/hour $\times$ 2.5 hours
 - * Distance = 150 miles
 2. In 1 hour, the train travels 60 miles
 - * In 2 hours, it's 120 miles
 - * In 0.5 hours, it's 30 miles
 - * Total: 120 + 30 = 150 miles
 3. Let's break it down:
 - * 1 hour: 60 miles
 - * 1 hour: 60 miles
 - * 0.5 hour: 30 miles
 - * Sum: 60 + 60 + 30 = 150 miles
 - Cross-Evaluation: All three solutions are consistent and mutually supportive. They use different approaches but arrive at the same result.
 - Resonance Graph: Fully connected graph with high edge weights
 - Resonance Score: 0.95
 - Calibrated Uncertainty: 0.05
 - Final Output: The train will travel 150 miles in 2.5 hours. (Uncertainty: 0.05)
 - Explanation: SRUQ generates multiple solution paths, evaluates their consistency, and quantifies uncertainty based on their semantic resonance. The high resonance score indicates low uncertainty, which is then calibrated to provide a final uncertainty estimate.

6. Fallback Plan: If SRUQ does not significantly outperform baselines, we can pivot to an analysis paper exploring why semantic resonance might not capture uncertainty effectively. We could investigate the quality and diversity of generated solution attempts, potentially improving the prompting strategies. Additionally, we could examine the effectiveness of the cross-evaluation step, possibly incorporating external knowledge or more structured reasoning. Furthermore, we could explore the relationship between graph properties and actual uncertainty, which might reveal insights about how LLMs represent confidence internally. We could also consider combining SRUQ with traditional uncertainty quantification methods, creating a hybrid approach that leverages both statistical and semantic information.